

Dan Sapp

Senior Geospatial Software Engineer

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[LinkedIn](#) | [GitHub](#) | [Interactive Engineering Portfolio](#)

Senior Geospatial Software Engineer focused on building interactive mapping and operational intelligence platforms using React, TypeScript, OpenLayers, Mapbox, Cesium, and PostGIS. Experienced developing mission-critical geospatial systems for defense and search-and-rescue operations, including real-time situational awareness tools, geospatial annotation workflows, AI-assisted command interfaces, and spatial intelligence applications.

SKILLS

- Geospatial: Mapbox GL, OpenLayers, Cesium/Resium, Raster/Vector Tiles, PostGIS, GeoJSON
- Frontend: TypeScript, React.js, Next.js, Tailwind, Redux, RTK Query, PWA, WebGL
- Backend: Node.js, FastAPI (Python), PostgreSQL, Supabase
- Infrastructure: GitHub Actions, Vercel, Render

PROFESSIONAL EXPERIENCE

Senior Geospatial Software Engineer | EarthDaily Federal – Geospatial Defense Contractor | 03/2025 – 02/2026 (1 Year Contract)

- Led frontend development of a geospatial search-and-rescue platform for the Air Force Rescue Coordination Center, replacing a paper-based workflow for distress beacon triage and mission management, improving speed and clarity in time-critical rescue operations. The system was iteratively refined through direct operator feedback during live beta deployment, strengthening coordination across multiple agencies.
- Architected an AI-assisted command interface using React and FastAPI that translated natural language prompts into structured geospatial UI actions through a dynamic tool registry.
- Built a Mapbox-based geospatial intelligence interface in collaboration with Army analysts, designed to integrate machine learning models for change detection and object detection (e.g., airstrips and vessel movement) and OSINT data tied to areas of interest.
- Developed an interactive geospatial annotation workspace supporting map labeling, callouts, and composable overlays, including drag-and-drop widgets such as overview maps and picture-in-picture views for comparative analysis.
- Proposed and prototyped a report generation workflow that was adopted into the product roadmap, enabling analysts to compile images and intelligence artifacts from map and annotation tools into structured reports with live PDF preview and in-place editing.
- Government Security Clearance
- Tech Stack: React, TypeScript, OpenLayers, Mapbox GL, FastAPI (Python)

Senior Software Engineer | Stellar/Storable – Marina Management Software | 02/2021 – 05/2024

- Developed multi-tenant web applications using React and TypeScript, supporting booking, sales, and customer management workflows across separate PostgreSQL-backed systems.
- Integrated third-party payment providers and handled sensitive data in compliance with PCI requirements while improving UI consistency across applications.

Senior Software Engineer | Flowhub – Dispensary Management Software | 07/2019 – 11/2020

- Built a responsive POS and inventory management interface using React and Redux for dispensary operations, focusing on usability and performance in high-transaction workflows.
- Developed a discount optimization service using Node.js and Web Workers to calculate optimal cart pricing combinations, improving upsell opportunities at checkout.

Software Engineer | Enterprise & Early Career Experience | 2015 – 2019

- Built React, Angular, and legacy JavaScript applications across healthcare, retail e-commerce, and restaurant POS systems, supporting enrollment workflows, product and cart experiences, and distributed device-based inventory operations.
- Developed a WebSocket-based cloud POS and inventory system enabling real-time communication across distributed restaurant devices using AngularJS.
- Collaborated with UX and engineering teams to improve usability, performance, and delivery velocity across multiple production systems, including insurance enrollment and retail platforms such as BestBuy.com.

EDUCATION

B.S. Mathematical Physics – Saint Cloud State University

A.S. Biomedical Technology – Anoka-Ramsey Community College